

Supply Chain → Order → Complete Information Architecture

1. Order Core Information

The basic identity and structure of the order.

- Order ID
- Order number
- Purchase order (PO) number
- Sales order number
- Internal reference number
- Marketplace order ID
- Channel source
- Order creation date
- Order confirmation date
- Order status
- Order priority level
- Order tags/labels
- Order type
- B2B
- B2C
- Wholesale
- Retail
- Subscription
- Pre-order
- Backorder
- Replacement
- Return/reverse order
- Parent-child order relationships
- Split order information
- Merged order information
- Order version history
- Revision logs
- Workflow stage
- SLA target

2. Customer Information

Everything related to the buyer.

- Customer ID

- Customer name
 - Company name
 - Customer segment
 - Customer tier
 - Email
 - Phone number
 - Alternate contact
 - Billing address
 - Shipping address
 - Geo-location
 - Delivery preferences
 - Preferred language
 - Tax ID/GSTIN/VAT
 - Customer lifetime value
 - Purchase history
 - Credit limit
 - Payment terms
 - Fraud score
 - Customer support history
 - Account manager
 - Loyalty points
 - Subscription details
 - Customer notes
 - Risk profile
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3. Product Information

Data related to products inside the order.

- SKU
- Product ID
- UPC/EAN/barcode
- Product name
- Brand
- Category
- Variant
- Size
- Color
- Material
- Unit of measure
- Quantity ordered
- Quantity allocated
- Quantity shipped
- Quantity delivered
- Quantity returned

- Quantity cancelled
 - Batch/lot number
 - Serial number
 - Expiry date
 - Manufacturing date
 - Hazard classification
 - Product dimensions
 - Product weight
 - Packaging type
 - Storage conditions
 - Country of origin
 - Compliance certifications
 - Product images
 - BOM (bill of materials)
 - Product substitution mapping
 - Bundle/kit relationships
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4. Pricing & Financial Information

All money-related order intelligence.

- Product price
- Cost price
- Landed cost
- Margin
- Gross profit
- Net profit
- Discount amount
- Coupon/promo applied
- Tax amount
- Currency
- Currency conversion rate
- Freight charges
- Insurance charges
- Handling charges
- Platform commission
- Marketplace fees
- COD charges
- Refund amount
- Return cost
- Payment gateway fee
- Payment reconciliation status
- Invoice number
- Invoice status
- Credit memo

- Debit memo
 - Financial approval workflow
 - Budget allocation
 - Revenue recognition status
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5. Payment Information

Order payment lifecycle.

- Payment status
 - Payment method
 - Transaction ID
 - Authorization status
 - Capture status
 - Settlement status
 - Refund status
 - Chargeback information
 - EMI/installment details
 - Wallet information
 - UPI details
 - Bank details
 - Payment gateway logs
 - Failed payment reasons
 - Retry attempts
 - Fraud detection results
 - Payment timestamps
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6. Inventory Information

Inventory intelligence connected to orders.

- Inventory availability
- Available-to-promise (ATP)
- Safety stock level
- Reorder point
- Current warehouse stock
- Reserved stock
- Damaged stock
- Transit stock
- Inbound inventory
- Outbound inventory
- Inventory aging
- Dead stock

- Fast-moving inventory
 - Slow-moving inventory
 - Multi-warehouse allocation
 - Warehouse transfer requirement
 - Supplier lead time
 - Inventory forecast
 - Inventory turnover ratio
 - Real-time stock sync
 - Stock discrepancy alerts
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7. Warehouse Information

Warehouse operations intelligence.

- Warehouse ID
 - Warehouse location
 - Warehouse capacity
 - Rack/bin location
 - Pick list
 - Packing list
 - Picker assigned
 - Packer assigned
 - Picking status
 - Packing status
 - Putaway status
 - Dock assignment
 - Shift allocation
 - Warehouse labor performance
 - Scan events
 - RFID tracking
 - Automation system logs
 - Robotic fulfillment data
 - Temperature monitoring
 - Security logs
 - Damage reports
 - Warehouse SLA compliance
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8. Fulfillment Information

Everything related to fulfillment execution.

- Fulfillment center
- Fulfillment status

- Allocation logic
- Fulfillment priority
- Partial fulfillment status
- Backorder handling
- Order batching
- Wave planning
- Pick-pack-ship cycle time
- Packing material used
- Gift wrapping
- Special handling instructions
- Quality inspection status
- Fulfillment exceptions
- Fulfillment automation logs
- Fulfillment cost
- Labor cost
- Processing time
- Order throughput

9. Shipping & Logistics Information

Transportation and delivery intelligence.

- Shipment ID
- Tracking number
- Carrier name
- Shipping method
- Delivery partner
- Dispatch date
- Estimated delivery date
- Actual delivery date
- Transit status
- Route optimization
- Last-mile delivery status
- Driver information
- Vehicle information
- Fuel cost
- Shipping label
- Shipping zone
- Delivery attempts
- Failed delivery reason
- POD (proof of delivery)
- Delivery photo
- Signature confirmation
- Geofencing events
- Customs information

- International shipping compliance
 - Freight forwarding details
 - Container tracking
 - Airway bill
 - Bill of lading
 - Transportation mode
 - Air
 - Sea
 - Rail
 - Road
 - Cold-chain monitoring
 - Real-time GPS tracking
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10. Supplier & Procurement Information

Supplier-side order intelligence.

- Supplier ID
 - Supplier name
 - Vendor rating
 - Vendor SLA
 - Procurement status
 - RFQ details
 - Quotation comparison
 - Purchase request
 - Purchase approval workflow
 - Supplier lead time
 - Supplier inventory visibility
 - MOQ (minimum order quantity)
 - Contract details
 - Procurement cost
 - Supplier risk score
 - Compliance documents
 - Supplier payment terms
 - Supplier communication logs
 - Procurement forecast
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11. Manufacturing Information

For production-linked supply chains.

- Production order ID
- Work order status

- Manufacturing stage
 - Assembly line status
 - Machine utilization
 - Downtime logs
 - Production capacity
 - Yield rate
 - Scrap rate
 - Quality defects
 - Production cycle time
 - Labor productivity
 - Raw material availability
 - WIP (work in progress)
 - Production forecast
 - Factory location
 - Shift performance
 - Maintenance schedule
 - IoT sensor data
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12. Return & Reverse Logistics Information

Everything after delivery.

- Return request ID
 - Return reason
 - Return status
 - Pickup scheduling
 - Reverse shipment tracking
 - Refund status
 - Exchange status
 - Restocking status
 - Return inspection results
 - Damage assessment
 - Warranty claims
 - Repair status
 - Refurbishment details
 - Disposal status
 - Return fraud detection
 - Return cost analysis
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13. Analytics & KPI Information

Business intelligence layer.

- Order conversion rate
 - Fill rate
 - Perfect order rate
 - OTIF (on-time in-full)
 - Order cycle time
 - Inventory turnover
 - Demand forecast accuracy
 - Cancellation rate
 - Return rate
 - Warehouse efficiency
 - Transportation efficiency
 - Supplier performance score
 - Cost per order
 - Revenue per order
 - Average order value
 - Customer acquisition cost
 - Customer retention rate
 - Gross merchandise value
 - Profitability analysis
 - SLA breach percentage
 - Real-time dashboards
 - Predictive analytics
 - AI recommendations
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14. AI/Automation Intelligence

Advanced supply chain intelligence.

- Demand prediction
- Auto replenishment
- Dynamic pricing
- Smart inventory allocation
- AI route optimization
- Fraud prediction
- Customer churn prediction
- Supplier risk prediction
- Delay prediction
- Inventory anomaly detection
- Automated exception handling
- AI procurement suggestions

- Predictive maintenance
 - Computer vision quality checks
 - Autonomous warehouse coordination
 - Intelligent order splitting
 - AI customer support agent logs
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15. Compliance & Legal Information

Regulatory and compliance layer.

- Tax compliance
 - Import/export compliance
 - Customs documentation
 - Dangerous goods compliance
 - Sustainability tracking
 - ESG metrics
 - Audit logs
 - Access logs
 - Regulatory certifications
 - GDPR/privacy compliance
 - Contract records
 - Insurance documentation
 - Legal disputes
 - Recall management
 - Traceability records
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16. Communication & Collaboration Information

Operational communication layer.

- Internal notes
- Customer communication
- Supplier communication
- Chat logs
- Email history
- Approval comments
- Escalation logs
- Notifications
- Alerts
- Task assignments
- Workflow approvals
- Collaboration threads
- Incident reports

- Resolution history
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17. Real-Time Operational Events

Live operational event stream.

- Order placed
 - Order updated
 - Payment received
 - Inventory allocated
 - Picking started
 - Packing completed
 - Shipment dispatched
 - Shipment delayed
 - Out for delivery
 - Delivered
 - Returned
 - Refunded
 - Exception raised
 - SLA breached
 - Inventory low alert
 - Supplier delay alert
 - Warehouse congestion alert
 - Weather disruption alert
 - Customs hold alert
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18. Advanced Enterprise Layer

What advanced global companies track.

- Digital twin simulation
- Supply chain graph/network mapping
- Multi-echelon inventory optimization
- Carbon footprint per order
- Supplier geopolitical risk
- Commodity price impact
- Demand sensing
- Scenario planning
- Capacity simulation
- Global trade lane analytics
- AI control tower visibility
- Real-time operational twin
- Autonomous procurement agents

- Autonomous warehouse orchestration
 - Autonomous transport planning
 - Enterprise knowledge graph
 - Unified operational memory
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19. Startup Opportunity Layer

What can become powerful startup products.

A. AI Supply Chain Control Tower

Single unified intelligence dashboard for entire operations.

B. Predictive Delay Engine

Predict delays before they happen.

C. Autonomous Procurement Agent

AI negotiates and places supplier orders automatically.

D. Unified Multi-Channel Order Brain

One system connecting Shopify, Amazon, Flipkart, ERP, warehouse, logistics.

E. AI Inventory Optimization System

Prevents overstock and stockouts.

F. Supply Chain Digital Twin

Simulate future operations before execution.

G. AI Exception Resolution Engine

AI automatically handles operational issues.

H. Live Supply Chain Graph Intelligence

Visual operational network of suppliers, warehouses, transport, and customers.

I. Autonomous Fulfillment Optimization

AI continuously improves packing, routing, warehouse flow.

J. Executive Decision Intelligence Layer

CEO-level operational intelligence with predictive insights.